Introducing WSOD-SAM Proposals and Heuristic Pseudo-Fully Supervised Training Strategy for Weakly Supervised Object Detection in Remote Sensing Images

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***Abstract*—Weakly supervised object detection (WSOD) in remote sensing images (RSI) relies solely on image-level labels, thereby significantly reducing the cost of instance-level annotations. Recently, pseudo-fully supervised object detection (PFSOD) models trained with image-level labels have achieved superior performance compared to traditional WSOD models. However, two major challenges remain. First, the pseudo ground truth (PGT) used in existing PFSOD models often suffers from low localization accuracy, resulting in suboptimal detection results. Second, existing PFSOD models tend to miss detections when multiple objects of the same category exist within an RSI. To address these issues, we propose a novel PFSOD framework incorporating two key innovations. To handle the first challenge, a WSOD-guided segment anything model (WSOD-SAM) is introduced to generate object proposals instead of traditional selective search (SS) algorithms based on low-level features. The WSOD-SAM proposals can combine the detection capability of the WSOD model with the segmentation capability of segment anything model (SAM); therefore, the baseline WSOD model equipped with them can produce more accurate PGTs. To handle the second challenge, we design a heuristic pseudo-fully supervised training (HPFST) strategy, which dynamically mines potential objects with high class confidence scores as pseudo soft labels for the classification branch of the YOLO model, in addition to using the PGT. Ablation studies confirm the individual and combined effectiveness of WSOD-SAM proposals and HPFST strategy. Extensive experiments demonstrate that our method achieves 78.12% and 31.44% mAP on the NWPU VHR-10.v2 and DIOR datasets, respectively, outperforming 16 existing WSOD methods. Our code is available at https://github.com/ccu6/hps-yolo**

***Index Terms*—Heuristic pseudo-fully supervised training (HPFST), pseudo-fully supervised object detection (PFSOD), remote sensing image (RSI), weakly supervised object detection (WSOD), WSOD-guided segment anything model (WSOD-SAM).**

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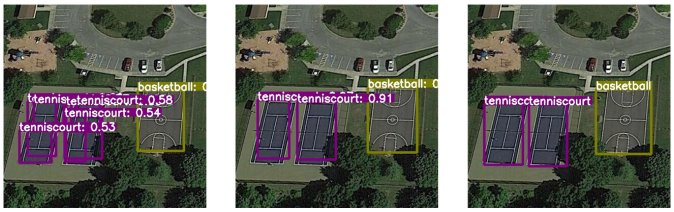
I. Introduction

OBJECT detection in remote sensing images (RSIs) is widely used in land surveying, urban planning, and military reconnaissance. So far, fully supervised object detection (FSOD) models [1], [2], [3], [4], [5], [6] and other related research areas [7], [8], [9], [10], [11], [12] have achieved excellent performance. However, the training of FSOD models requires instance-level annotations, which demand enormous labor and time costs. In contrast, weakly supervised object detection (WSOD) models, which only require image-level annotations, have attracted significant research interest in recent years [13], [14] Current WSOD models are broadly classified into two categories: traditional WSOD and pseudo-FSOD (PFSOD) models. A pioneering traditional WSOD model, WSDDN [13], adopts a selective search (SS) algorithm [15] to generate multiple object proposals and employs multiple instance learning (MIL) to produce the class confidence score (CCS) of each proposal. A milestone PFSOD model, SoS-WSOD [16], uses the detection results of a traditional WSOD model to produce pseudo ground truth (PGT), which is then used to train an FSOD model through a teacher–student network. Benefiting from the rapid development of fully supervised training techniques, PFSOD models have achieved superior performance compared to [16], [17], and [18] and have become a research hotspot. However, existing PFSOD models still suffer from two major problems, which are detailed below.

The first problem is the insufficient localization accuracy of the PGT used in current PFSOD models. As mentioned earlier, PGTs are derived from the detection results of traditional WSOD models, whose localization accuracy mainly depends on the quality of object proposals. Traditional WSOD models usually adopt the SS algorithm to generate object proposals; however, the SS algorithm relies on low-level features lacking semantic information, leading to limited localization accuracy.

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(a) (b) (c)

Fig. 1. Visualized detection results of (a) baseline model and (b) baseline model with WSOD-SAM proposals, and (c) ground truth.



(a) (b) (c)

Fig. 2. Visualized detection results of (a) traditional PFSOD model and (b) PFSOD model trained using the HPFST strategy, and (c) ground truth.

As shown in Fig. 1(a), the baseline model that adopts SS proposals fails to accurately locate the tennis courts.

The second problem is that missed detections easily occur in current PFSOD models when multiple objects of the same category exist in an RSI. Specifically, traditional WSOD models usually focus on the most salient object rather than all objects of the same category, since each image-level label can be determined by any single object of the corresponding category. Consequently, the PGT derived from traditional WSOD models cannot fully cover all objects, and the PFSOD models trained with such incomplete PGTs tend to miss detections. As illustrated in Fig. 2(a), the traditional PFSOD model fails to detect all airplanes.

To address the first problem, we propose a WSOD-guided segment anything model (WSOD-SAM) to generate object proposals instead of using SS, and the resulting proposals are referred to as WSOD-SAM proposals in this article. The core idea of WSOD-SAM is to leverage the detection results of a traditional WSOD model to generate prompt points that guide the segment anything model (SAM) for more accurate object segmentation. The final object proposals are then derived from the segmentation masks. Though existing work [19] has attained promising object proposals based on SAM, however, the quality of proposals is limited by directly leveraging SAM without any prior knowledge. As a matter of fact, the SAM tends to split the entire object into multiple fragmented small regions under the condition of lacking object prompts. In contrast, our approach effectively combines the object detection capability of the WSOD model with the segmentation capability of SAM, resulting in high-quality object proposals. As illustrated in Fig. 1(b), the baseline model with WSOD-SAM proposals accurately locates the tennis courts.

To address the second problem, we propose a heuristic pseudo-fully supervised training (HPFST) strategy for training PFSOD models. The core idea of the HPFST strategy is online mining potential objects from the detection results of PFSOD model, serving as the auxiliary supervision of PFSOD model besides the PGT generated by traditional WSOD models, and the potential objects are continuously updated during training. As shown in Fig. 2(b), the PFSOD model trained using the HPFST strategy successfully detects all of airplanes.

In addition, this article develops an improved baseline WSOD model based on OICR [20], which integrates the seed instance mining strategy from OD-WSCL [21] and the invariant multiple instance learning (IMIL) scheme from CICR [14]. Compared with the original OICR, our baseline model achieves a remarkable performance improvement.

The main contributions of this work are summarized as follows.

1) We propose the WSOD-SAM proposals and replace traditional SS proposals in WSOD models. By combining the detection capability of WSOD with the segmentation power of SAM, it effectively enhances the localization accuracy of the PGT derived from WSOD models.
2) We propose an HPFST strategy that can be applied to any PFSOD model. It dynamically mines potential objects from model predictions as auxiliary supervision, thereby alleviating the missed detection problem when multiple instances of the same category appear in an RSI.
3) We construct an enhanced WSOD baseline by integrating the OICR framework with the seed instance mining strategy from OD-WSCL [21] and the IMIL scheme from CICR [14]. This improved baseline achieves significant gains over the original OICR model.

II. Related Works

A. Traditional WSOD Models

Traditional WSOD models utilize MIL and image-level labels to derive the initial CCS for each object proposal. They then mine seed instances to optimize multiple instance classifier refinement (ICR) branches. Finally, the CCS of each object proposal is jointly inferred from all ICR branches. According to the seed instance mining strategy, traditional WSOD models can be divided into two categories.

The first category of models solely relies on the CCS of each proposal to identify seed instances. OICR [20] adopts WSDDN [13] to obtain the initial CCS of each proposal and adds multiple ICR branches to progressively refine these CCSs. In OICR, the proposal with the highest CCS is selected as the seed instance and combined with its neighboring proposals to produce the PGT for the subsequent ICR branch. MIST [22] builds upon the OICR framework by selecting proposals with high CCS as initial seed instances and refining them through non-maximum suppression. Furthermore, a regression branch is attached to each ICR branch to enhance localization accuracy, supervised by the seed instances generated from the previous ICR branch. Other representative works include PCL [23], TCA [24], PCIR [25], and RINET [26].

The second category of models leverages both the CCS of each proposal and additional cues to mine seed instances.

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OD-WSCL [21] utilizes the proposal with the highest CCS and its neighboring proposals to construct a category-specific feature bank. It then computes the similarity between each proposal's feature and the corresponding feature bank, and proposals with similarity scores exceeding a predefined threshold are selected as seed instances of that category. MHG-PSL [27] computes a segmentation-based dual-context projection score to assess proposal completeness, and the weighted sum of the projection score and CCS is used to identify seed instances. Other notable works in this category include MIG [28], MELM [29], and CICR [14].

B. PFSOD Models

PFSOD models utilize the prediction results of traditional WSOD models to generate PGT, which is then used to train an FSOD model serving as the final detector. According to the fully supervised training strategy, existing PFSOD models can be divided into two categories.

The first category adopts a single-pass training scheme. SGPLM-IR [30] combines weakly supervised semantic segmentation masks with CCSs to mine high-quality seed instances, yielding more reliable detection results. These results are then converted into PGT through an enhanced PGT instance generation strategy and used to train an FSOD head. WeakSAM [17] employs the SAM to improve the performance of WSOD models and uses its predictions as PGT to train an FSOD model, where PGT instances with training losses exceeding a predefined threshold are discarded during training.

The second category of PFSOD models employs an iterative training strategy. SoS-WSOD [16] divides the prediction results of WSOD models into two subsets: labeled PGT with smaller training losses and unlabeled PGT with larger losses. The labeled and unlabeled PGTs are jointly used to train a teacher–student network iteratively, where the teacher network generates pseudo-labels for the unlabeled PGT, which are only used to update the classification branch. IPS-YOLO [18] proposes a CCS-based weighted synthesis scheme to generate the initial PGT from the output of an improved OICR model. The PGT quality is then continuously refined through an iterative updating strategy, while a YOLO detector is trained on the refined PGT. The trained YOLO detector subsequently produces the final detection results.

III. Proposed Method

A. Overview

As illustrated in Fig. 3, the training procedure of the proposed method consists of three stages. In the first stage, a traditional MIL-based baseline model is constructed, and an initial weakly supervised training of it is conducted. In the second stage, multiple WSOD-SAM proposals are generated by combining the SAM with the detection results of the baseline model. The weakly supervised training of baseline model is conducted again, guided by the WSOD-SAM proposals to enhance localization accuracy. In the third stage, a filtering operation is applied to refine the initial PGT generated by the trained baseline model. The refined PGT is subsequently used to train a YOLO detector via the proposed HPFST strategy. The resulting YOLO model serves as the final detector, producing the ultimate detection results.

B. Initial Weakly Supervised Training of Baseline Model

1) Baseline Model:

a) Feature extraction and MIL head: As illustrated in Fig. 3(a) and (b), a training RSI is first fed into the backbone network to extract feature maps, while the SS algorithm is employed to generate object proposals $P = \{P_1, \ldots, P_n, \ldots, P_N\}$, where N denotes the number of proposals. In this article, P is referred to as SS proposals. Subsequently, a region of interest (RoI) pooling operation is applied to the feature maps of the RSI to obtain the feature maps of each proposal. These proposal-level feature maps are then passed through two cascaded fully connected (FC) layers to generate the feature vectors of proposals.

The feature vectors are subsequently input into an MIL head to predict the image-level CCS of the RSI. Specifically, the feature vectors are first passed through an FC layer to generate a classification matrix M_{cls} and a detection matrix M_{det}. These matrices are then combined to produce the CCS matrix of P, denoted as $P_{\text{CCS}} \in \mathbb{R}^{(C\times N)}$, which is computed as

$$P_{\text{CCS}} = \sigma_c\left(M_{\text{cls}}\right) \odot \sigma_d\left(M_{\text{det}}\right) \tag{1}$$

where $\sigma_c(\cdot)$ and $\sigma_d(\cdot)$ denote the softmax operations over categories and proposals, respectively, $\odot$ represents element-wise multiplication, and C is the number of object categories. The image-level CCS of the RSI in the cth category, denoted as S^c_{RSI}, is computed as

$$S^c_{\text{RSI}} = \sum_{n=1}^{N} P^{c,n}_{\text{CCS}}, \quad c \in [1, C] \tag{2}$$

where $P^{c,n}_{\text{CCS}}$ denotes the CCS of P_n in cth category.

Finally, the binary cross-entropy (BCE) loss of the MIL head, denoted as L_M, is formulated as

$$L_M = -\sum_{c=1}^{C}\left(b_c \log S^c_{\text{RSI}} + (1-b_c)\log\left(1 - S^c_{\text{RSI}}\right)\right) \tag{3}$$

where b_c is a ground-truth binary indicator, i.e., $b_c = 1$ if the RSI contains an object of the cth category, and $b_c = 0$ otherwise.

b) IDR head based on OD strategy and IMIL scheme: The widely used OICR [20] introduces multiple ICR branches to predict the category of each proposal. However, the accumulated errors across ICR branches may result in misclassification, since the supervision signal for each ICR branch is not derived from the true labels [14]. To address this issue, inspired by the IMIL scheme [14], K instance detection refinement (IDR) heads are constructed to predict both the category and offsets of each proposal more accurately. Each IDR head consists of an ICR branch, an auxiliary branch, an MIL branch, and a regression branch. The details are described as follows.

The kth ICR branch utilizes the feature vectors of proposals to predict the CCS of proposals, with $k \in [1, K]$. The set

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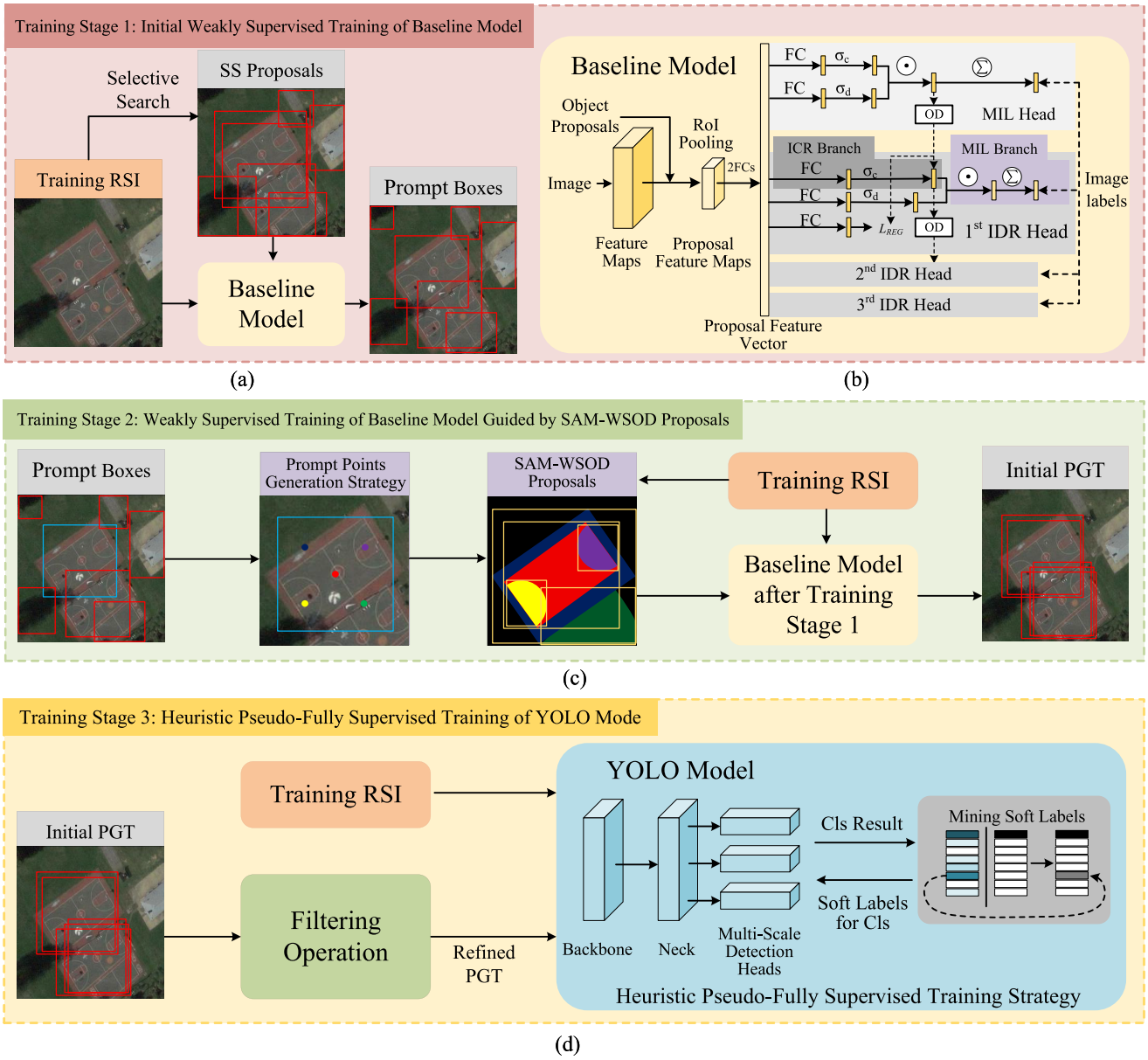


Fig. 3. Illustration of the proposed method. (a)–(d) Framework of training stage 1, baseline model, training stage 2, and training stage 3, respectively.

of positive samples in the cth category for the kth ICR branch, denoted as PS_c^k, consists of seed instances and their neighboring proposals, where the seed instances are mined using the OD strategy [21] in this work. Accordingly, the loss of the kth ICR branch, denoted as L_{ICR}^k, is formulated as follows:

$$L_{\mathrm{ICR}}^k = -\frac{1}{N}\sum_{n=1}^{N}\sum_{c=1}^{C+1} w_n^k b_{c,n}^k \log x_{c,n}^k \tag{4}$$

where $w_n^k = x_{c,n_c^{k-1}}^{k-1}$ denotes the weight of P_n in the kth ICR branch. Here, $x_{c,n_c^{k-1}}^{k-1}$ represents the CCS of proposal $P_{n_c^{k-1}}$ in the cth category of the $(k-1)$th ICR branch, and n_c^{k-1} indicates the index of the proposal with the highest CCS in the cth category of the $(k-1)$th ICR branch. $x_{c,n}^k$ denotes the CCS of P_n in the cth category of the kth ICR branch. The binary indicator $b_{c,n}^k = 1$ if $P_n \in \mathrm{PS}_c^k$, and $b_{c,n}^k = 0$ otherwise.

To mitigate the accumulated prediction errors of the K ICR branches, the IMIL scheme employs the ground-truth

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image-level labels to indirectly supervise all K branches [14]. First, the feature vectors of proposals are fed into the kth auxiliary branch to obtain the auxiliary score matrix $\mathrm{AS}^k \in \mathbb{R}^{(C+1)\times N}$, where each auxiliary branch comprises an FC layer followed by a σ_d operation. Next, the image-level CCS of the kth MIL branch can be computed as follows:

$$\begin{aligned} \mathrm{SP}^k &= \mathrm{AS}^k \odot X^k \\ \mathrm{SI}_c^k &= \sum_{n=1}^{N} \mathrm{SP}_{c,n}^k \end{aligned} \tag{5}$$

where X^k represents the CCS matrix output from the kth ICR branch, SP^k denotes the instance-level CCS matrix in the kth MIL branch, $\mathrm{SP}_{c,n}^k \in \mathrm{SP}^k$ indicates the CCS of P_n in the cth category and kth MIL branch, and SI_c^k denotes the image-level CCS of the kth MIL branch in the cth category.

Finally, the BCE loss of the kth MIL branch, denoted as L_{MIL}^k, is defined as

$$L_{\mathrm{MIL}}^k = -\sum_{c=1}^{C} \left(b_c \log \mathrm{SI}_c^k + (1-b_c) \log\left(1-\mathrm{SI}_c^k\right)\right). \tag{6}$$

In this way, the kth ICR branch is indirectly optimized through L_{MIL}^k.

Recent studies have reported that attaching a regression branch to each ICR branch can further enhance the localization accuracy of WSOD models [22], [27]. The loss of the kth regression branch, denoted as L_{REG}^k, is formulated as follows:

$$L_{\mathrm{REG}}^k = -\frac{1}{|\mathrm{PS}^k|} \sum_{z=1}^{|\mathrm{PS}^k|} \left(\mathrm{smooth}_{L1}\left(b_z, \widehat{b_z}\right)\right) \tag{7}$$

where $\mathrm{PS}^k = \{\mathrm{PS}_1^k, \ldots, \mathrm{PS}_c^k, \ldots, \mathrm{PS}_C^k\}$ denotes the set of positive samples in the kth ICR branch, $|\mathrm{PS}^k|$ is the number of positive samples, $\mathrm{smooth}_{L1}(\cdot)$ represents the Smooth L1 function [31], and b_z and $\widehat{b_z}$ indicate the predicted and ground-truth offsets of the zth positive sample, respectively.

2) Loss of Baseline Model: The loss of baseline model, denoted as L_B, is defined as follows:

$$L_B = L_M + \sum_{k=1}^{K} L_{\mathrm{IDR}}^k \tag{8}$$

where the loss of the kth IDR head, L_{IDR}^k, is given by

$$L_{\mathrm{IDR}}^k = L_{\mathrm{ICR}}^k + L_{\mathrm{MIL}}^k + L_{\mathrm{REG}}^k \tag{9}$$

where L_{IDR}^k represents the combined loss of the kth ICR branch, MIL branch, and regression branch. After training, the baseline model utilizes all K ICR and regression branches jointly to infer the detection results of the input training RSIs.

C. Weakly Supervised Training of Baseline Model-Guided by WSOD-SAM Proposals

1) Generating Prompt Boxes: Each training RSI is imported into the baseline model to obtain its detection results after the first training stage, and each detection result is considered as a prompt box of SAM, denoted as $\mathrm{pb}_m \in \mathrm{PB}$, $m \in [1, M]$, where PB denotes the set of prompt boxes, and $M = |\mathrm{PB}|$ denotes the number of prompt boxes.

Each training RSI is fed into the baseline model to obtain its detection results after the first training stage. Each detection result is treated as a prompt box for SAM, denoted as $\mathrm{pb}_m \in \mathrm{PB}, m \in [1, M]$, where PB is the set of prompt boxes and $M = |\mathrm{PB}|$ denotes the number of prompt boxes.

2) Generating Prompt Points: As illustrated in Fig. 3(c), each prompt box pb_m generates five prompt points. Considering the inconsistent quality of prompt boxes generated by WSOD model, directly adopting four vertices of these boxes as prompt points will induce the segmentation regions produced by these prompt points to contain substantial background regions during subsequent point-based segmentation. To alleviate the above problem, our solution is defined as follows:

$$\begin{aligned} x_m^1 &= x_m, \quad y_m^1 = y_m \\ x_m^2 &= x_m - \frac{w}{4}, \quad y_m^2 = y_m + \frac{h}{4} \\ x_m^3 &= x_m + \frac{w}{4}, \quad y_m^3 = y_m + \frac{h}{4} \\ x_m^4 &= x_m - \frac{w}{4}, \quad y_m^4 = y_m - \frac{h}{4} \\ x_m^5 &= x_m + \frac{w}{4}, \quad y_m^5 = y_m - \frac{h}{4} \end{aligned} \tag{10}$$

where (x_m, y_m) is the center coordinate of pb_m, and w and h denote the width and height of pb_m, respectively. The $(x_m^1, y_m^1), \ldots, (x_m^5, y_m^5)$ denote the coordinates of center, top-left, top-right, bottom-left, and bottom-right prompt points of pb_m, respectively.

3) Generating WSOD-SAM Proposals: As shown in Fig. 3(c), each prompt point guides SAM to generate a segment region, formulated as

$$\mathrm{sr}_m^l = \mathrm{SAM}\left(I_T, x_m^l, y_m^l\right), \quad l \in \{1,2,3,4,5\} \tag{11}$$

where I_T denotes the input training RSI, (x_m^l, y_m^l) denotes the coordinate of the lth prompt point of pb_m, $\mathrm{SAM}(\cdot)$ represents the segmentation operation conducted by the SAM, and sr_m^l denotes the segment region corresponding to (x_m^l, y_m^l).

The segment region sr_m^l is subsequently used to generate a WSOD-SAM proposal, denoted as WP_m^l, with spatial coordinates defined as

$$\begin{aligned} x_{\mathrm{TL},m}^l &= x_{\min}\left(\mathrm{sr}_m^l\right), \quad y_{\mathrm{TL},m}^l = y_{\max}\left(\mathrm{sr}_m^l\right) \\ x_{\mathrm{BR},m}^l &= x_{\max}\left(\mathrm{sr}_m^l\right), \quad y_{\mathrm{BR},m}^l = y_{\min}\left(\mathrm{sr}_m^l\right) \end{aligned} \tag{12}$$

where $x_{\min}(\cdot)$ and $x_{\max}(\cdot)$ denote finding minimum and maximum x-coordinates functions, and $y_{\min}(\cdot)$ and $y_{\max}(\cdot)$ denote finding minimum and maximum y-coordinates functions, respectively. $(x_{\mathrm{TL},m}^l, y_{\mathrm{TL},m}^l)$ and $(x_{\mathrm{BR},m}^l, y_{\mathrm{BR},m}^l)$ represent the top-left and bottom-right coordinates of WP_m^l, respectively.

4) Second-Stage Training of Baseline Model: All WSOD-SAM proposals are treated as object proposals and fed into the baseline model. The baseline model is then retrained under weak supervision, with the loss function described in Section III-B2.

D. HPFST of Yolo Model

1) Refinement of Initial PGT: Each training RSI is fed into the baseline model to obtain detection results after the second

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training stage. All detection results are considered as a set of initial PGT instances, denoted as IP.

Algorithm 1 Filtering Operation

Input: $IP = \{IP_1, \ldots, IP_c, \ldots, IP_C\}$ ▷IP_c denotes the initial PGT instances of c-th category
T_o ▷Overlap threshold
R_s ▷Shrink ratio of maximum CCS
$IS = \{IS_1, \ldots, IS_c, \ldots, IS_C\}$ ▷IS_c denotes the CCSs of IP_c, IS denotes the set of CCSs corresponding to IP
Output: $RP = \{RP_1, \ldots, RP_c, \ldots, RP_C\}$ ▷RP_c denotes the refined PGT instances of c-th category
Initialize $RP = \emptyset$
$T_c = max(IS\{:\}) \times Rs$ ▷$IS\{:\}$ denotes all of elements in $IS\{:\}$, T_c denotes the CCS threshold for IP
for $c = 1$ to C **do**
for $i = 1$ to $|IS_c|$ **do**
if $is_c^i \leq T_c$ **then** ▷ $is_c^i \in IS_c$ denotes the CCS of ip_c^i, $ip_c^i \in IP_c$ denotes the i-th initial PGT instance in IP_c
remove ip_c^i from IP_c
end if
end for
while $IP_c \neq \emptyset$ **do**
$im = \underset{j}{argmax}\,(is_c^j)$
for $i = 1$ *to* $|IP_c|$ **do**
$overlap = \frac{area(ip_c^{im} \cap ip_c^i)}{area(ip_c^{im})}$
if $overlap \geq T_o$ **then**
remove ip_c^i from IP_c
end if
end for
$RP_c = RP_c \cup ip_c^{im}$
end while
end for

A filtering operation is then applied to IP to obtain a set of refined PGT instances, denoted as RP. Details of the filtering operation are provided in Algorithm 1.

2) HPFST Strategy: The refined PGT instances RP along with the proposed HPFST strategy are employed to train a YOLO model. The detailed procedure of the HPFST strategy is provided in Algorithm 2.

As shown in Algorithm 2, the predicted detection results with high CCS, excluding the objects annotated in RP, are regarded as potential objects. These potential objects are employed to compute the classification loss via their soft labels. Consequently, this approach helps alleviate the missed detection problem described in Section I.

Considering that the potential objects are derived from the predicted detection results of YOLO model, the location accuracy of potential objects cannot be guaranteed. Therefore, the potential objects are only used for classification loss and not for regression loss.

Algorithm 2 HPFST Strategy

Input: RP
I_T ▷Input training RSI
T_s ▷CCS threshold
$YOLO^I$ ▷Initial YOLO model
Output: $YOLO^T$ ▷Trained YOLO model
for $e = 1$ to E **do** ▷E denotes the total number of training epochs
$(R, SR) = YOLO^I(I_T)$ ▷R denotes the predicted detection results of I_T, SR denotes the CCSs of R
$(LC, LR) = TA(R, SR, RP)$ ▷LC denotes the assigned one-hot classification labels of R, LR denotes the location labels used for the regression of R, $TA(\cdot)$ denotes the task-aligned label assigning strategy commonly used by YOLO model [32], [33]
if $e \geq 2$ **then**
for $c = 1$ to C **do**
for $d = 1$ to $|R_c|$ **do** ▷ $R_c \in R$ denotes the predicted detection results of c-th category in R
if $sr_c^d \geq T_s$ **then** ▷ $sr_c^d \in SR$ denotes the CCS of $r_c^d \in R_c$ which denotes the d-th predicted detection result in R_c
$lc_c^d = sr_c^d$ ▷lc_c^d denotes the soft label of r_c^d
$LC = LC \cup lc_c^d$
end if
end for
end for
end if
Calculating the classification and regression loss of R according to LC and LR, respectively, and using the sum of classification and regression loss to train $YOLO^I$
end for
$YOLO^T \leftarrow YOLO^I$

IV. Experiments

A. Dataset and Evaluation Metrics

Experiments are conducted on the NWPU VHR-10.v2 [34], [35] and DIOR [36] datasets.

The NWPU VHR-10.v2 dataset contains 2775 instances across 1172 RSIs with a resolution of 400 × 400 pixels. It includes 10 object categories: airplane, ship, storage tank, baseball diamond, tennis court, basketball court, ground track field, harbor, bridge, and vehicle. The numbers of training, validation, and testing samples are 679, 200, and 293, respectively.

The DIOR dataset consists of 192 472 instances across 23 463 RSIs with a resolution of 800 × 800 pixels. It spans 20 object categories: airplane, airport, baseball diamond, basketball court, bridge, chimney, dam, expressway toll station, expressway service area, golf course, ground track field, harbor, overpass, ship, stadium, storage tank, tennis court, train station, vehicle, and windmill. The numbers of training and testing samples are 11 725 and 11 738, respectively. For both datasets, the training and validation samples are used for model training.

The mean average precision (mAP) is employed to evaluate detection accuracy on the testing subset, while correct localization (CorLoc) is used to assess localization performance on the training and validation subsets.

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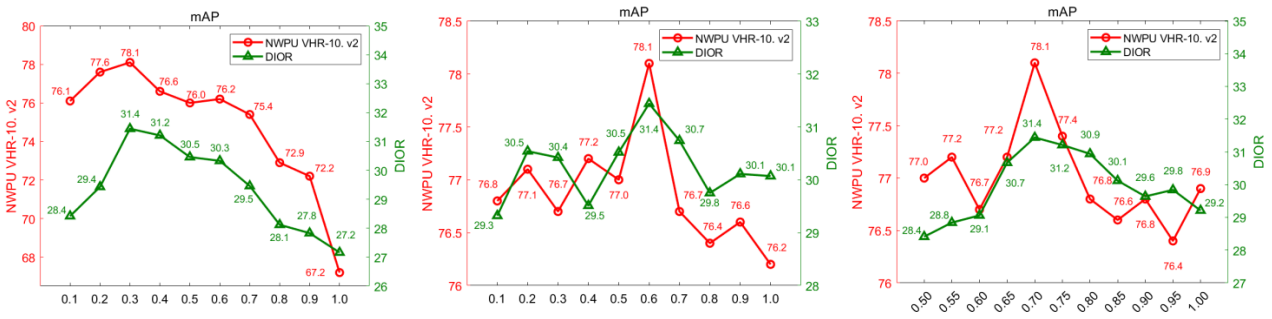


Fig. 4. Quantitative parameter analysis of (a) R_s, (b) T_o, and (c) T_s.

B. Implementation Details

1) Configuration of First and Second Training Stages: The VGG16 [44] network pretrained on ImageNet is used as the backbone of the baseline model. The baseline model is equipped with three IDR heads. The number of training iterations for the baseline model is 30–60 K on the NWPU VHR-10.v2 and DIOR datasets, respectively. The initial learning rate is set to 0.1 and is decreased by a factor of 10 at 20 K (50 K) and 26 K (56 K) iterations, and the batch size is set to 16 (8) for NWPU VHR-10.v2 (DIOR) dataset. Stochastic gradient descent (SGD) is adopted as the optimizer, with a momentum of 0.9 and a weight decay of 0.05.

2) Configuration of Third Training Stage and Inference Stage: YOLOv12m [45] is employed in the third training stage, and the final trained model is denoted as HPS-YOLO. The training configuration in the third stage follows the default YOLOv12 settings. In the inference stage of HPS-YOLO, the confidence score threshold is set to 0.25, and the IoU threshold of NMS is set to 0.7.

3) Environment Setting: The training stage is conducted on a multi-GPU server equipped with two Xeon E5-2560v4 CPUs and 512 GB of RAM. The first and second training stages are performed on four TITAN RTX GPUs (24 GB $\times$4 memory), with PyTorch 1.7.1 and CUDA 11.8. While the third training stage is conducted on a single TITAN RTX GPU (24 GB memory), with PyTorch 1.13 and CUDA 11.8.

C. Parameter Analysis

The parameters R_S and T_o from Algorithm 1, and T_s from Algorithm 2, are quantitatively analyzed on the NWPU VHR-10.V2 and DIOR datasets. As shown in Fig. 4, the mAP of HPS-YOLO achieves its highest value on both datasets when $R_S = 0.3$, $T_o = 0.6$, and $T_s = 0.7$.

D. Ablation Study

1) Ablation Studies of OD Strategy and IMIL Scheme: As shown in Table I, the mAP of OICR baseline model increases by 21.84% and 17.95% when integrating the OD strategy and the IMIL scheme independently, respectively. When both strategies are utilized simultaneously, the mAP is further improved by 30.73%, showing the effectiveness of the OD strategy, the IMIL scheme, and their combination.

TABLE I
ABLATION STUDY OF OD STRATEGY AND IMIL SCHEME ON THE NWPU VHR-10.V2 DATASET

OICR Baseline	OD Strategy	IMIL Scheme	mAP(%)
✓			34.52
✓	✓		56.36
✓		✓	52.47
✓	✓	✓	65.25

TABLE II
ABLATION STUDY OF WSOD-SAM PROPOSALS AND HPFST STRATEGY ON THE NWPU VHR-10.V2 AND DIOR DATASETS

Baseline	SAM-WSOD Proposals	HPFST Strategy	mAP(%)	
			NWPU VHR-10.V2	DIOR
✓			65.25	20.32
✓	✓		68.99	24.57
✓		✓	72.23	27.19
✓	✓	✓	78.12	31.44

2) Ablation Studies of WSOD-SAM Proposals and HPFST Strategy: As shown in Table II, the mAP of the baseline model increases by 3.74% (4.25%) and 7.02% (6.87%) when using the WSOD-SAM proposals and the HPFST strategy independently on the NWPU VHR-10.V2 (DIOR) datasets, respectively. When both strategies are employed simultaneously, the mAP is further improved by 12.81% (11.12%) on the NWPU VHR-10.V2 (DIOR) datasets, demonstrating the effectiveness of the WSOD-SAM proposals, the HPFST strategy, and their combination.

E. Comparison With Popular WSOD Methods on the NWPU VHR-10.v2 Dataset

As shown in Tables III and IV, compared with 16 existing WSOD methods, including WSDDN [13], OICR [20], PCL [23], MIST [22], DCL [37], PCIR [25], MIG [28], TCA [24], SAE [38], SPG [39], MHQ-PSL [27], SGPLM-IR [30], PIIFR [40], HIMWR [41], RINet [26], AE-IS [42], and IENet [43], the proposed HPS-YOLO model achieves significant improvements in both mAP and CorLoc. Specifically, the mAP (CorLoc) of HPS-YOLO is

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TABLE III

COMPARISON WITH POPULAR METHODS ON THE NWPU VHR-10.V2 DATASET IN TERMS OF MAP (%)

Method	Airplane	Ship	Storage tank	Baseball Diamond	Tennis Court	Basketball Court	Ground Track field	Harbor	Bridge	Vehicle	mAP
WSDDN [13]	30.08	41.72	34.98	88.90	12.86	23.85	99.43	13.94	1.92	3.60	35.12
OICR [20]	13.66	67.35	57.16	55.16	13.64	39.66	92.80	0.23	1.84	3.73	34.52
PCL [23]	26.00	63.76	2.50	89.80	64.45	76.07	77.94	0.00	1.30	15.67	39.41
MIST [22]	69.68	49.16	48.55	80.91	27.08	79.85	91.34	46.99	8.29	13.36	51.52
DCL [37]	72.70	74.25	37.05	82.64	36.88	42.27	83.95	39.57	16.82	35.00	52.11
PCIR [25]	90.78	78.81	36.40	90.80	22.64	52.16	88.51	42.36	11.74	35.49	54.97
MIG [28]	88.69	71.61	75.17	94.19	37.45	47.68	100.00	27.27	8.33	9.06	55.95
TCA [24]	89.43	78.18	78.42	90.80	35.27	50.36	90.91	42.44	4.11	28.30	58.82
SAE [38]	82.91	74.47	50.20	96.74	55.66	72.94	100.00	36.46	6.33	31.89	60.72
SPG [39]	90.42	81.00	59.53	92.31	35.64	51.44	99.92	58.71	16.99	42.99	62.80
MHQ-PSL [27]	87.60	81.00	57.30	94.00	36.40	80.40	100.00	56.90	9.80	35.60	63.80
SGPLM-IR [30]	90.70	79.90	69.30	97.50	41.60	77.50	100.00	44.40	17.20	33.50	65.20
PIIFR [40]	90.80	81.60	56.60	91.70	51.90	69.50	100.00	53.40	16.30	40.50	65.20
HIMWR [41]	90.81	80.52	73.42	96.59	47.35	78.94	100.00	43.89	18.35	33.51	66.24
RINet [26]	90.30	86.30	79.60	90.70	58.20	80.40	100.00	57.70	18.90	41.60	70.40
AE-IS [42]	91.00	88.20	78.30	93.20	60.60	82.40	100.00	60.40	19.60	45.80	72.00
IENet [43]	—	—	—	—	—	—	—	—	—	—	72.90
HPS-YOLO (Ours)	99.50	67.82	78.85	97.07	87.16	97.91	84.37	92.97	0.27	75.30	78.12

TABLE IV

COMPARISON WITH POPULAR METHODS ON THE NWPU VHR-10.V2 DATASET IN TERMS OF CORLOC (%)

Method	Airplane	Ship	Storage tank	Baseball Diamond	Tennis Court	Basketball Court	Ground Track field	Harbor	Bridge	Vehicle	CorLoc
WSDDN [13]	22.32	36.81	39.95	92.48	17.96	24.24	99.26	14.83	1.69	2.89	35.24
OICR [20]	29.41	83.33	20.51	81.76	40.85	32.08	86.60	7.41	3.70	14.44	40.01
PCL [23]	11.76	50.00	12.82	98.65	84.51	77.36	90.72	0.00	9.26	15.56	45.06
MIST [22]	90.20	82.50	80.30	98.60	48.50	87.40	98.30	66.50	14.60	35.80	70.30
PCIR [25]	100.00	93.06	64.10	99.32	64.79	79.25	89.69	62.96	13.26	52.22	71.87
MIG [28]	97.79	90.26	87.18	98.65	54.93	64.15	100.00	74.07	12.96	21.57	70.16
TCA [24]	96.91	91.78	95.13	88.65	66.90	62.88	95.98	54.18	19.63	55.50	72.76
SAE [38]	97.06	91.67	87.81	98.65	40.86	81.13	100.00	70.37	14.81	52.22	73.46
SPG [39]	98.06	92.67	70.08	99.65	51.86	80.12	96.20	72.44	12.99	60.02	73.41
MHQ-PSL [27]	94.40	86.60	68.50	97.80	69.80	87.50	100.00	68.60	16.00	56.60	74.30
SGPLM-IR [30]	98.20	93.80	89.30	99.10	50.20	88.90	100.00	71.00	12.30	51.20	75.40
PIIFR [40]	98.63	94.70	76.45	83.92	61.39	82.47	100.00	78.16	19.55	57.24	75.20
HIMWR [41]	99.29	94.79	91.68	97.95	58.43	91.67	100.00	68.78	13.64	56.68	76.89
AE-IS [42]	98.30	94.20	72.40	100.00	56.80	83.60	98.40	76.80	18.20	62.40	76.10
HPS-YOLO (Ours)	95.66	96.22	96.28	95.45	95.90	95.51	88.19	96.55	3.56	95.81	85.91

TABLE V

COMPARISON WITH POPULAR METHODS ON THE DIOR DATASET IN TERMS OF MAP (%)

Method	Airplane	Airport	Baseball field	Basketball court	Bridge	Chimney	Dam	Expressway service area	Expressway toll station	Golf field
WSDDN [13]	9.06	39.68	37.81	20.16	0.25	12.28	0.57	0.65	11.88	4.90
OICR [20]	8.70	28.26	44.05	18.22	1.30	20.15	0.09	0.65	29.89	13.80
PCL [23]	21.52	35.19	59.80	23.49	2.95	43.71	0.12	0.90	1.49	2.88
MIST [22]	32.01	39.87	62.71	28.97	7.46	12.87	0.31	5.14	17.38	51.02
DCL [37]	20.89	22.70	54.21	11.50	6.03	61.01	0.09	1.07	31.01	30.87
PCIR [25]	30.37	36.06	54.22	26.60	9.09	58.59	0.22	9.65	36.18	32.59
MIG [28]	22.20	52.57	62.76	25.78	8.47	67.42	0.66	8.85	28.71	57.28
TCA [24]	25.13	30.84	62.92	40.00	4.13	67.78	8.07	23.80	29.89	22.34
SAE [38]	20.57	62.41	62.65	23.54	7.59	64.62	0.22	34.52	30.62	55.38
SPG [39]	31.32	36.66	62.79	29.10	6.08	62.66	0.31	15.00	30.10	35.00
MHQ-PSL [27]	29.10	49.80	70.90	41.40	7.20	45.50	0.20	35.40	36.80	60.80
SGPLM-IR [30]	39.10	64.60	64.40	26.90	6.30	62.30	0.90	12.20	26.30	55.30
PIIFR [40]	32.90	70.50	63.20	45.70	0.20	69.70	0.20	12.40	39.40	56.40
HIMWR [41]	42.19	65.01	66.15	25.74	6.70	60.15	1.29	13.48	25.31	57.81
RINet [26]	26.20	57.40	62.70	25.10	9.90	69.20	1.40	13.30	36.20	51.40
AE-IS [42]	31.80	50.90	63.20	29.40	8.90	68.70	1.30	15.10	35.50	51.60
IENet [43]	—	—	—	—	—	—	—	—	—	—
HPS-YOLO (Ours)	70.56	35.51	71.27	34.53	2.34	51.14	2.33	0.04	41.76	0.36

Method	Ground track field	Harbor	Overpass	Ship	Stadium	Storage tank	Tennis court	Train station	Vehicle	Windmill	mAP
WSDDN [13]	42.53	4.66	1.06	0.70	63.03	3.95	6.06	0.51	4.55	1.14	13.26
OICR [20]	57.39	10.66	11.06	9.09	59.29	7.10	0.68	0.14	9.09	0.41	16.50
PCL [23]	56.36	16.76	11.05	9.09	57.62	9.09	2.47	0.12	4.55	4.55	18.19
MIST [22]	49.48	5.36	12.24	29.43	35.53	25.36	0.81	4.59	22.22	0.80	22.18
DCL [37]	56.45	5.05	2.65	9.09	63.65	9.09	10.36	0.02	7.27	0.79	20.19
PCIR [25]	58.51	8.60	21.63	12.09	64.28	9.09	13.62	0.30	9.09	7.52	24.92
MIG [28]	47.73	23.77	0.77	6.42	54.13	13.15	4.12	14.76	0.23	2.43	25.11
TCA [24]	58.85	24.84	11.06	9.09	46.40	13.74	30.98	1.47	9.09	1.00	25.82
SAE [38]	52.70	17.57	6.85	9.09	51.59	15.43	1.69	14.44	1.41	9.16	27.10
SPG [39]	48.02	27.11	12.00	10.02	60.04	15.10	21.00	9.92	3.15	0.06	25.77
MHG-PSL [27]	48.50	14.00	25.10	18.50	48.90	11.70	11.90	3.50	11.30	1.70	28.60
SGPLM-IR [30]	60.60	9.40	23.10	13.40	57.40	17.70	1.50	14.00	11.50	3.50	28.50
PIIFR [40]	55.30	16.60	0.60	9.10	54.80	18.30	11.00	16.10	9.10	1.10	29.10
HIMWR [41]	61.42	10.41	20.40	14.14	58.62	18.91	2.16	13.61	10.02	4.69	28.91
RINet [26]	53.90	28.60	4.80	9.10	52.70	15.80	20.60	12.90	9.10	4.70	28.30
AE-IS [42]	52.30	28.80	13.30	11.20	56.90	16.30	22.40	14.00	8.00	2.60	29.10
IENet [43]	—	—	—	—	—	—	—	—	—	—	30.20
HPS-YOLO (Ours)	78.31	27.35	2.96	61.90	4.13	38.45	73.03	2.88	29.04	0.89	31.44

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TABLE VI
COMPARISON WITH POPULAR METHODS ON THE DIOR DATASET IN TERMS OF CORLOC (%)

Method	Airplane	Airport field	Baseball field	Basketball court	Bridge	Chimney	Dam	Expressway service area	Expressway toll station	Golffield	
WSDDN [13]	5.72	59.88	94.24	55.94	4.92	23.40	1.03	6.79	44.52	12.75	
OICR [20]	15.98	51.45	94.77	55.79	3.55	23.89	0.00	4.82	56.68	22.42	
PCL [23]	61.14	46.86	95.39	63.61	7.32	95.07	0.21	5.71	5.14	50.77	
MIST [22]	91.60	53.20	93.50	66.30	10.80	30.70	1.50	14.30	35.20	47.50	
PCIR [25]	93.10	45.60	95.50	68.30	3.60	92.10	0.20	5.40	58.40	47.50	
MIG [28]	76.98	46.86	95.39	63.61	23.00	95.07	0.21	16.96	57.88	50.77	
TCA [24]	81.58	51.33	96.17	73.45	5.03	94.69	15.89	32.79	45.95	48.56	
SAE [38]	91.20	69.37	95.48	67.52	18.88	97.78	0.21	70.54	54.32	51.43	
SPG [39]	80.48	32.04	98.68	65.00	15.20	96.08	22.52	16.99	46.08	50.96	
MHQ-PSL [27]	85.50	68.90	96.80	75.80	11.60	94.70	0.80	67.50	60.50	46.50	
SGPLM-IR [30]	92.20	58.30	97.80	74.20	16.20	95.20	0.30	51.30	56.20	52.30	
PIIFR [40]	88.30	69.10	98.80	69.40	19.9	97.80	0.30	24.70	56.20	54.40	
HIMWR [41]	94.14	59.34	98.12	70.46	17.59	94.61	4.80	52.19	54.27	53.47	
RINet [26]	92.70	80.90	92.70	69.50	8.60	90.10	0.20	71.30	62.0	65.5	
AE-IS [42]	91.40	78.60	96.10	68.80	16.00	92.30	22.80	68.90	60.60	62.70	
IENet [43]	—	—	—	—	—	—	—	—	—	—	
HPS-YOLO (Ours)	80.79	93.72	94.86	77.14	34.78	99.26	10.76	1.07	78.42	6.59	

Method	Ground track field	Harbor	Overpass	Ship	Stadium	Storage	Tennis court	Train station	Vehicle	Windmill	CorLoc
WSDDN [13]	89.90	5.45	10.00	22.96	98.54	79.61	15.06	3.45	11.56	3.22	32.44
OICR [20]	91.41	18.18	18.70	31.80	98.28	81.29	7.45	1.22	15.83	1.98	34.77
PCL [23]	89.39	42.12	19.78	37.94	97.93	80.65	13.77	0.20	10.50	6.94	41.52
MIST [22]	87.10	38.60	23.40	50.70	80.50	89.20	22.40	11.50	22.20	2.40	43.60
PCIR [25]	88.60	15.80	5.20	39.50	98.10	85.60	13.40	56.50	9.70	0.60	46.12
MIG [28]	89.39	42.12	19.78	37.94	97.93	80.65	13.77	10.34	10.50	6.94	46.80
TCA [24]	85.26	39.91	20.17	30.63	84.59	91.46	56.28	3.79	10.45	1.25	48.41
SAE [38]	88.28	48.03	2.28	33.56	14.11	83.35	65.59	19.88	16.41	2.85	49.42
SPG [39]	89.18	49.45	22.00	35.16	98.61	90.04	32.56	12.73	9.98	2.34	48.30
MHQ-PSL [27]	75.20	50.50	28.30	39.70	92.60	77.00	55.10	10.10	20.90	5.60	53.20
SGPLM-IR [30]	91.70	48.60	23.00	32.70	98.80	89.30	49.50	19.50	18.30	4.00	53.20
PIIFR [40]	89.60	49.10	19.40	34.50	96.70	84.70	63.20	15.60	11.60	3.40	52.30
HIMWR [41]	93.04	49.71	22.04	34.81	99.01	90.42	44.13	18.31	17.52	10.51	53.92
RINet [26]	85.10	51.40	15.70	44.60	98.60	80.30	14.80	22.70	6.90	2.60	52.80
AE-IS [42]	88.20	50.90	23.40	40.20	98.80	91.50	33.20	18.40	12.30	2.50	55.90
IENet [43]	—	—	—	—	—	—	—	—	—	—	56.10
HPS-YOLO (Ours)	95.96	42.56	25.54	94.78	19.10	90.84	93.60	19.88	70.50	18.71	57.44

improved by 42.98% (50.67%), 43.58% (45.9%), 38.69% (40.85%), 26.58% (15.61%), 26.58%, 23.13% (14.04%), 22.15% (15.75%), 19.28% (13.15%), 17.38% (12.45%), 15.3% (12.5%), 14.3% (11.61%), 12.9% (10.51%), 12.9% (10.71%), 11.86% (9.02%), 7.7% 6.1% (9.81%), and 5.22%, respectively.

F. Comparison With Popular WSOD Methods on the DIOR Dataset

The comparison with 16 WSOD methods is conducted on the large-scale DIOR dataset. As shown in Tables V and VI compared with WSDDN, OICR, PCL, MIST, DCL, PCIR, MIG, TCA, SAE, SPG, MHQ-PSL, SGPLM-IR, PIIFR, HIMMR, RINet AE-IS, and IENet, the proposed HPS-YOLO model achieves significant improvements in both mAP and CorLoc. Specifically, the mAP (CorLoc) of HPS-YOLO is improved by 18.14% (25.0%), 14.9% (22.67%), 13.21% (15.92%), 9.22% (13.84%), 11.21%, 6.48% (11.32%), 6.29% (10.64%), 5.58% (9.03%), 4.3% (8.02%), 5.63% (9.14%), 2.8% (4.24%), 2.9% (4.24%), 2.3% (5.14%), 2.49% (3.52%), 3.1% (4.64%), and 2.3% (1.54%), 1.24% (1.34%), respectively.

In summary, the above results demonstrate that the proposed HPS-YOLO exhibits superior detection and localization performance on both the NWPU VHR-10.V2 and DIOR datasets compared with 16 existing WSOD methods.

G. Evaluation of Computational Cost

To evaluate the computational cost of HPS-YOLO, it is compared with popular WSOD models in terms of parameters, GFLOPs, and FPS. As shown in Table VII, the HPS-YOLO is superior to six WSOD models across all metrics. In addition, three variants of HPS-YOLO are compared with each other, where the YOLOv12s, YOLOv12m, and YOLOv12l are adopted as the baseline models of HPS-YOLOs, HPS-YOLOm, and HPS-YOLOl, respectively. The mAP of HPS-YOLOm is apparently higher than that of HPS-YOLOs and is close to that of HPS-YOLOl, while the computational cost of HPS-YOLOm is significantly lower than that of HPS-YOLOl, giving the best comprehensive performance. Therefore, the YOLOv12m is selected as the baseline FSOD model of HPS-YOLO.

TABLE VII
COMPARISON WITH OTHER WSOD MODELS ON THE NWPU VHR-10.V2 DATASET IN TERMS OF COMPUTATIONAL COST

Method	Parameters(M)	GFLOPs	FPS	mAP (%)
OICR [20]	134.60	286.23	4.86	34.52
MIST [22]	135.02	290.43	4.62	51.52
RINet [26]	135.11	534.26	3.94	70.40
PISLM [30]	135.53	296.43	4.47	63.80
SGPLM-IR [27]	135.63	334.26	4.26	65.20
CICR [14]	135.08	294.37	4.50	73.10
HPS-YOLOs	9.3	21.4	242.16	72.37
HPS-YOLOm	20.1	67.2	133.32	78.12
HPS-YOLOl	26.4	88.9	102.04	78.19

H. Subjective Evaluation

To intuitively demonstrate the improvement of HPS-YOLO, a subjective comparison between HPS-YOLO and baseline model is presented in this section. As shown in Fig. 5(a), the HPS-YOLO locates storage tanks more precisely than the baseline model because of the proposed WSOD-SAM proposals. A similar case can be seen in Fig. 5(b). As shown in Fig. 5(c), a missed tennis court in the baseline model is

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Fig. 5. Subjective comparison between HPS-YOLO and baseline model. (a)–(d) are cited from the NWPU VHR-10.V2 dataset.



Fig. 6. Illustration of some failure detections. (a)–(d) are cited from the DIOR dataset.

correctly detected by the HPS-YOLO because of the proposed HPFST strategy. Similar cases can be seen in Fig. 5(b) and (d).

I. Analysis of Failure Detection Results

Although our HPS-YOLO model has certain advantages over other WSOD models in terms of detection accuracy and speed, it still has faults in certain scenarios. Fig. 6 shows some typical faults, and their reasons are summarized as follows.

1) Co-Existing Background: Some image-level labels are jointly determined by objects and their contextual regions, which are so-called co-existing backgrounds. Therefore, the co-existing backgrounds will be mistakenly detected by WSOD models when instance-level labels are absent. As shown in Fig. 6(a), the reservoir is the co-existing background of dam; therefore, it is detected by our baseline WSOD model. Consequently, the HPS-YOLO trained by the PGT derived from the baseline WSOD model also mistakenly detects the reservoir.

2) Object With Irregular Shape: The global shape of some object category is irregular; therefore, WSOD model can only discover the shared local characteristics. As shown in Fig. 6(b), the shape of golf field is irregular; consequently, only local fairways are detected by our model.

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3) Similar Background: Some background regions are similar to foreground objects in some scenarios; therefore, they are mistakenly detected as objects. As shown in Fig. 6(c), an auxiliary runway is mistakenly detected as airport.

4) Shadow Interference: The shadow of object is more salient than the object itself in some scenarios; therefore, WSOD model mistakenly detects the shadow rather than the object. As shown in Fig. 6(d), the shadows of windmills are mistakenly detected by our model.

V. Conclusion

In this article, we propose a novel PFSOD framework supervised by image-level labels, whose key innovations lie in the introduction of WSOD-SAM proposals and the HPFST strategy. First, the traditional SS proposals based on low-level features are replaced by the proposed WSOD-SAM proposals in the baseline model, thereby addressing the issue of insufficient localization accuracy in the PGT generated by conventional baselines. Second, the proposed HPFST strategy is adopted in the pseudo-supervised training of the YOLO model, which dynamically selects potential objects with high CCSs as pseudo-labels for the classification branch, in addition to using the PGT. This effectively alleviates the missed detection problem in current PFSOD models. Furthermore, an improved WSOD baseline model is constructed by incorporating OD-WSCL and CICR schemes into the OICR framework, achieving notable performance gains over the traditional OICR baseline. Ablation studies validate the effectiveness of the WSOD-SAM proposals and the HPFST strategy. Extensive experiments on the NWPU VHR-10.v2 and DIOR datasets demonstrate that our method achieves significant improvements in both detection and localization accuracy compared with 16 existing WSOD methods.

In future research, the proposed WSOD-SAM proposals and HPFST strategy can be applied to more advanced WSOD baseline models to obtain more accurate detection results. Moreover, integrating language-guided segmentation models may further enhance the quality of object proposals.

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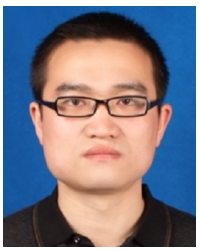
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